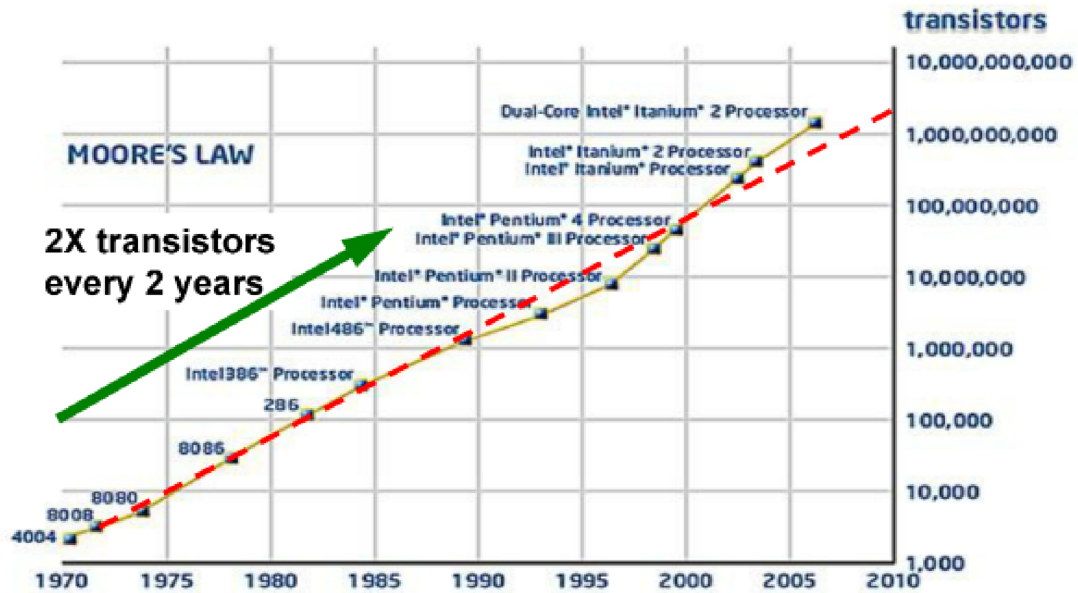
Nanotechnology

Silicon Nanotube Pattern Formed by Synchrotron Radiation Lithography

Dr. Terry Y. H. Tang

Background

40+ Years of Moore's Law at INTEL: From Few to Billions of Transistors

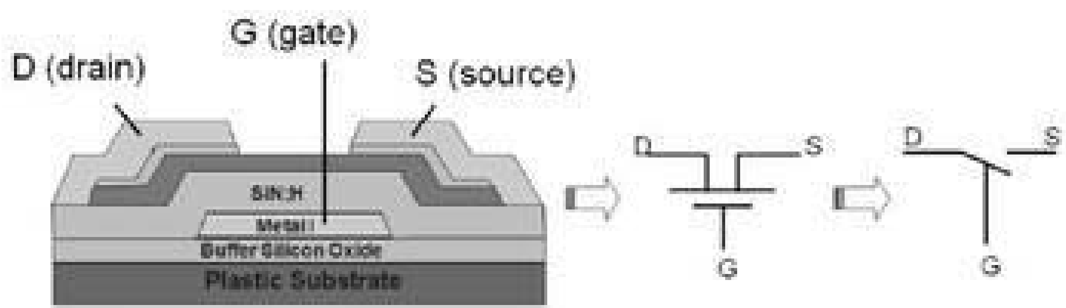


Transistor Count has Doubled Every Two Years

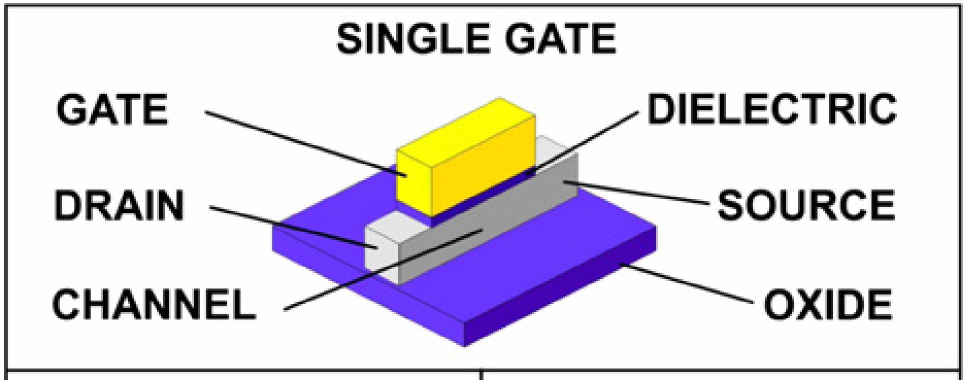

VISUALIZING PROGRESS
If transistors were people
If the transistors in a microprocessor were represented by people, the following timeline gives an idea of the pace of Moore's Law.
2,300
Average music hall capacity
134,000
Large stadium capacity
32 Million
Population of Tokyo
1.3 Billion
Population of China
1970
1980
1990
2000
2011
Intel 4004
Intel 286
Pentium III
Core i7 Extreme Edition

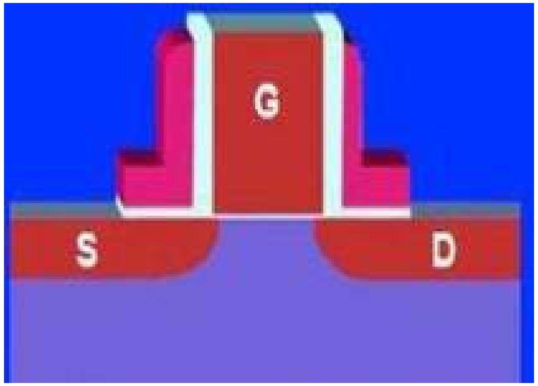

intel inside
pentium
EXTREME
EDITION


TECLAST
T7200
T090701 0027

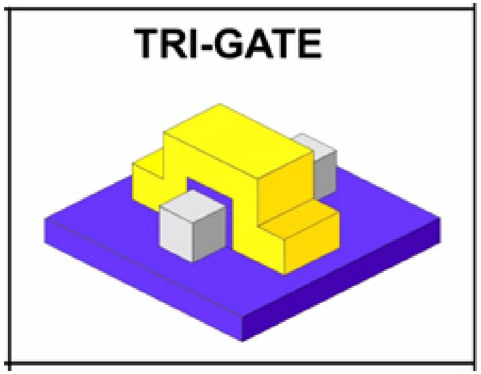

G (gate)
D (drain)
S (source)
SiN:H
Metal I
Buffer Silicon Oxide
Plastic Substrate
D
S
G
D
S
G

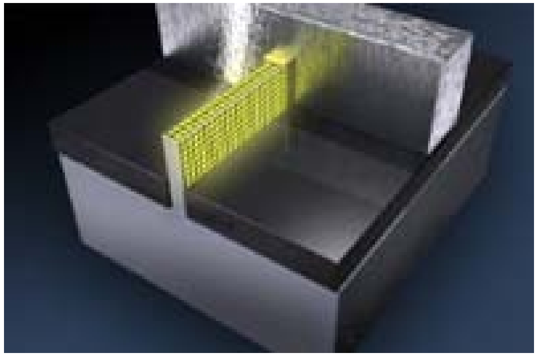
Old product

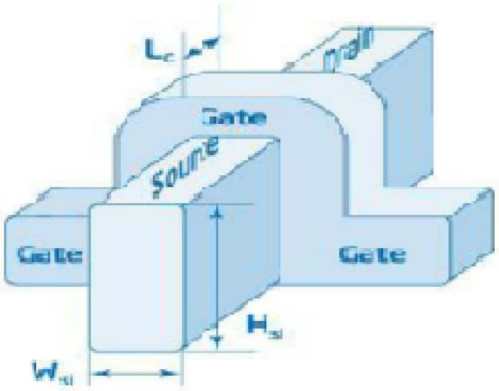

SINGLE GATE
GATE
DIELECTRIC
DRAIN
SOURCE
CHANNEL
OXIDE


G
S
D

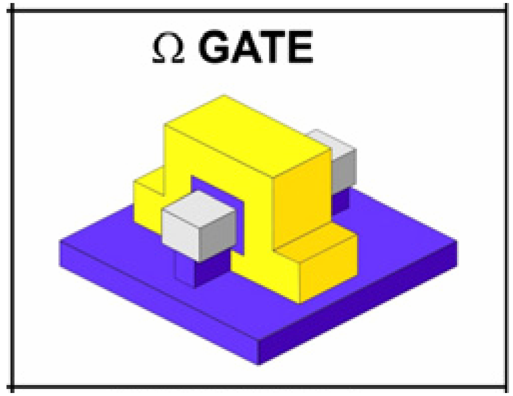
New Product : Intel® 22nm Technology (the world's first 3-D transistor)



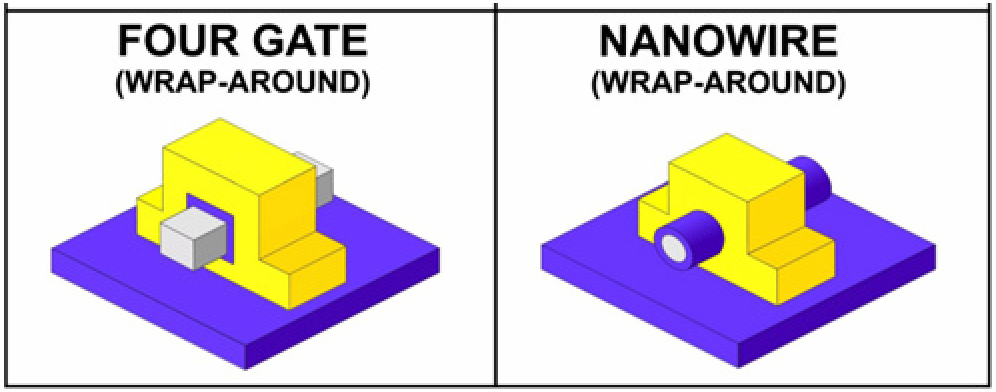




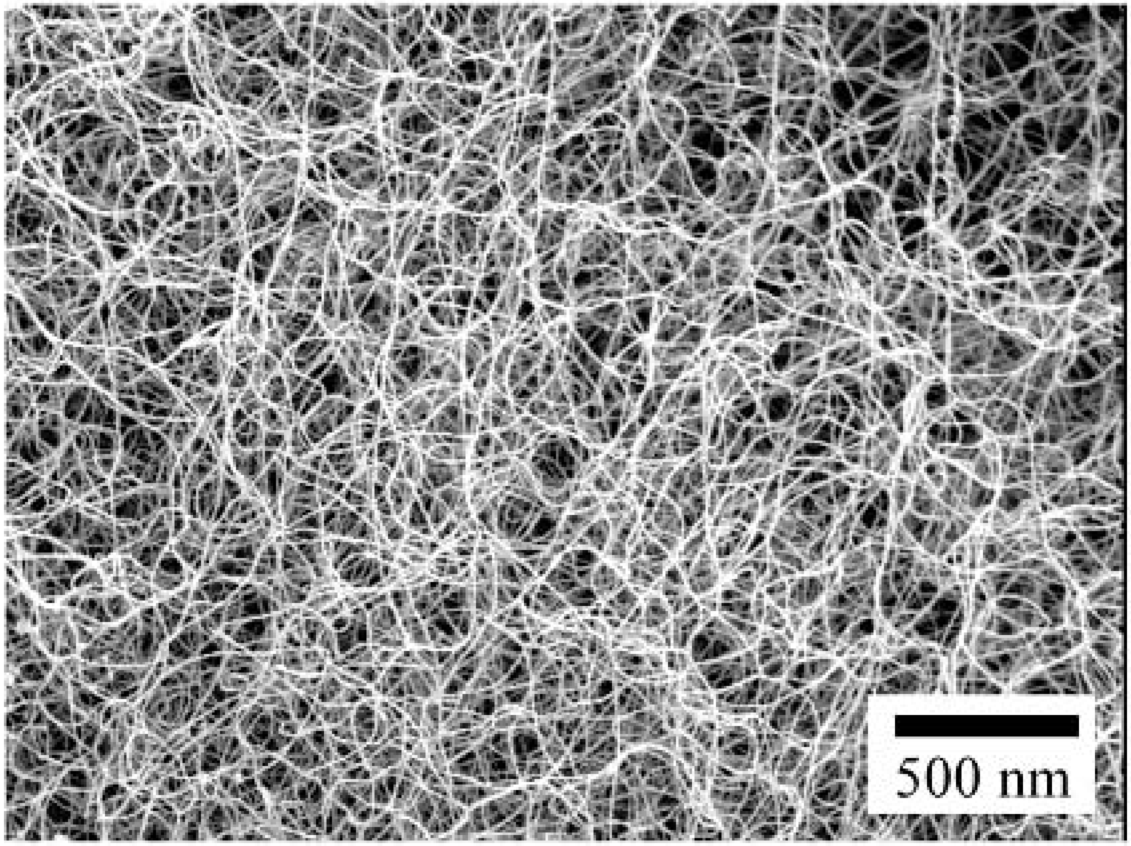
Now studied



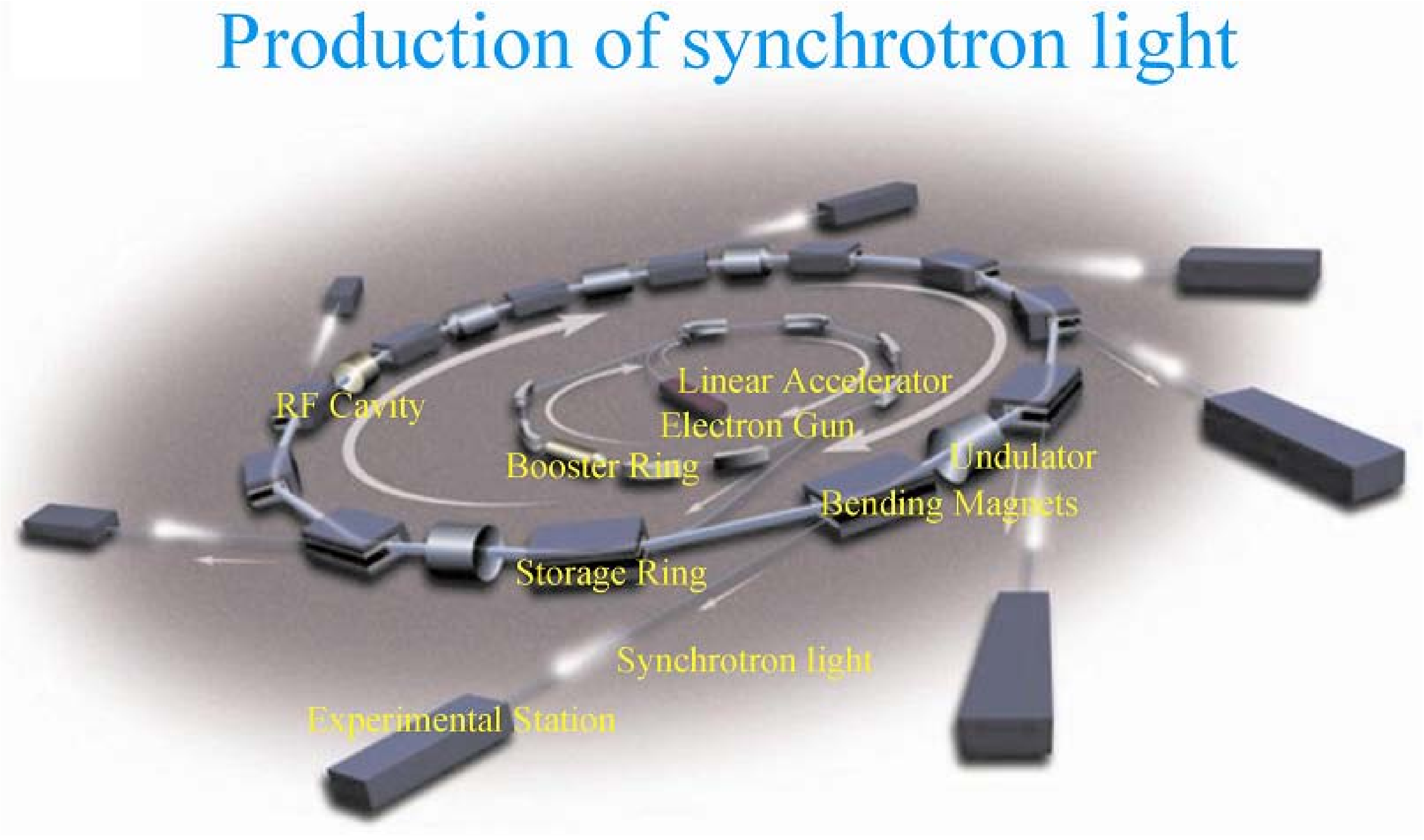
Dreamed 4-D
or A-D Gate



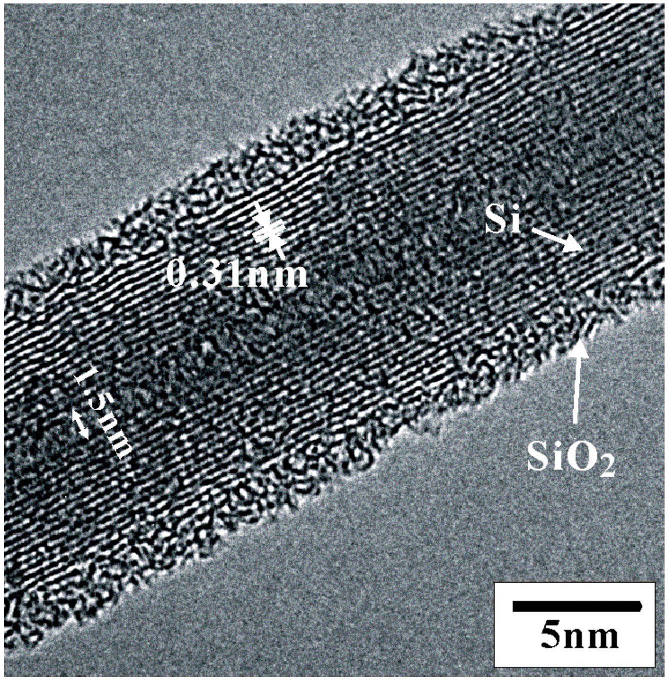
SEM image of Si nanowires

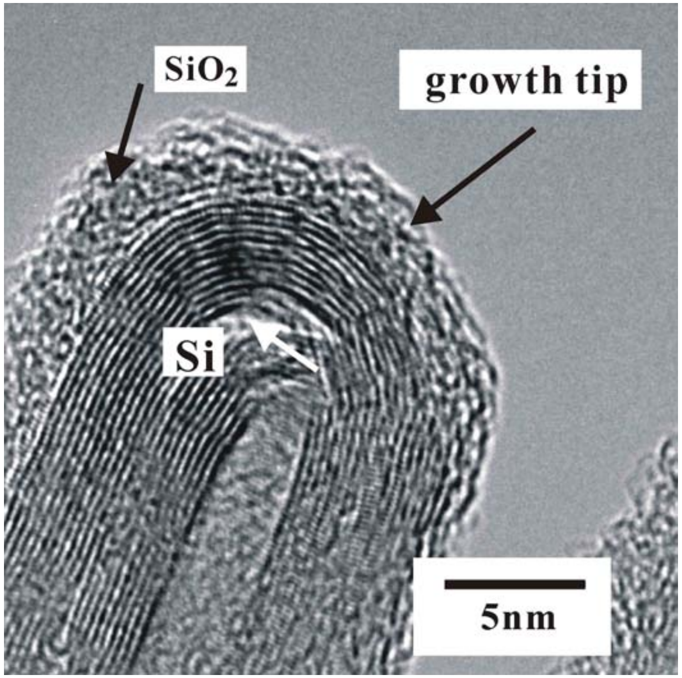


My Suggestion



Silicon nanotube or silicon hollow nanowire





PRL 95, 116102 (2005) PHYSICAL REVIEW LETTERS week ending 9 SEPTEMBER 2005

Self-Assembled Silicon Nanotubes under Supercritically Hydrothermal Conditions

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(Received 11 October 2004; published 7 September 2005)

Self-assembled silicon nanotubes with one-dimensional structure have been synthesized from silicon monoxide powder under supercritically hydrothermal conditions with a temperature of 470 °C and a pressure of 6.8 MPa. The silicon nanotubes were identified by transmission electron microscopy and high-resolution transmission electron microscopy. The results show that the silicon nanotubes (SiNT) have closed caps. The structures of the silicon nanotubes are hollow inner pore, crystalline silicon wall layers with a 0.31 nm interplanar spacing and 2–3 nm amorphous silica outer layers. Pure crystalline silicon nanotubes survive after etching the silicon nanotubes with 5% HF acid for enough time to imply that the self-assembled silicon nanotubes are stable. A possible theoretical reason for the growth of SiNTs from SiO under supercritically hydrothermal conditions was also proposed.

DOI: 10.1103/PhysRevLett.95.116102 PACS numbers: 81.07.De, 61.46.+w


The United States of America

The Director of the United States Patent and Trademark Office

Has received an application for a patent for a new and useful invention. The title and description of the invention are enclosed. The requirements of law have been complied with, and it has been determined that a patent on the invention shall be granted under the law.

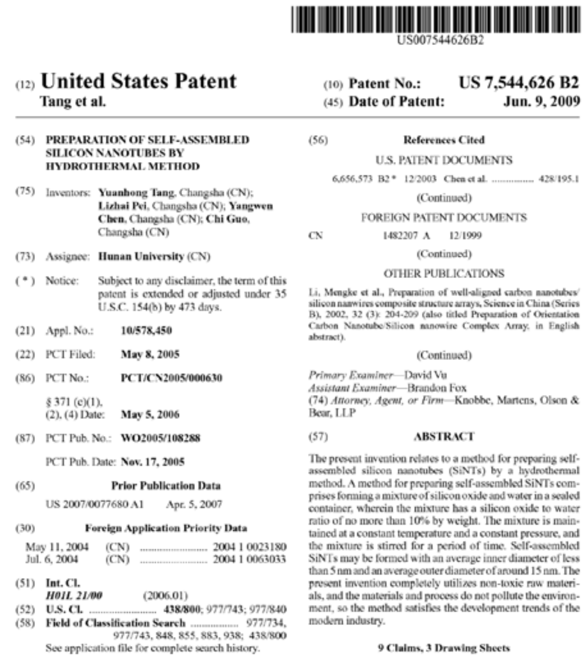
Therefore, this

United States Patent

Grants to the person(s) having title to this patent the right to exclude others from making, using, offering for sale, or selling the invention throughout the United States of America or importing the invention into the United States of America for the term set forth below, subject to the payment of maintenance fees as provided by law.

If this application was filed prior to June 8, 1995, the term of this patent is the longer of seventeen years from the date of grant of this patent or twenty years from the earliest effective U.S. filing date of the application, subject to any statutory extension.

If this application was filed on or after June 8, 1995, the term of this patent is twenty years from the U.S. filing date, subject to any statutory extension. If the application contains a specific reference to an earlier filed application or applications under 35 U.S.C. 120, 121 or 365(c), the term of the patent is twenty years from the date on which the earliest application was filed, subject to any statutory extensions.


US007544626B2

(12) **United States Patent**
Tang et al.

(10) **Patent No.:** **US 7,544,626 B2**
(45) **Date of Patent:** **Jun. 9, 2009**

(54) **PREPARATION OF SELF-ASSEMBLED SILICON NANOTUBES BY HYDROTHERMAL METHOD**

(75) Inventors: **Yuanhong Tang**, Changsha (CN); **Lizhai Pei**, Changsha (CN); **Yangwen Chen**, Changsha (CN); **Chi Guo**, Changsha (CN)

(73) Assignee: **Hunan University** (CN)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 473 days.

(21) Appl. No.: **10/578,450**

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(30) **Foreign Application Priority Data**

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Jul. 6, 2004 (CN) 2004 1 0063033

(51) **Int. Cl.**
H01L 21/00 (2006.01)
(52) **U.S. Cl.** **438/800**; 977/743; 977/840
(58) **Field of Classification Search** 977/734, 977/743, 848, 855, 883, 938; 438/800
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

6,656,573 B2 * 12/2003 Chen et al. 428/195.1

(Continued)

FOREIGN PATENT DOCUMENTS

CN 1482207 A 12/1999

(Continued)

OTHER PUBLICATIONS

Li, Mengke et al., Preparation of well-aligned carbon nanotubes/silicon nanwires composite structure arrays, Science in China (Series B), 2002, 32 (3): 204-209 (also titled Preparation of Orientation Carbon Nanotube/Silicon nanowire Complex Array, in English abstract).

(Continued)

Primary Examiner—David Vu
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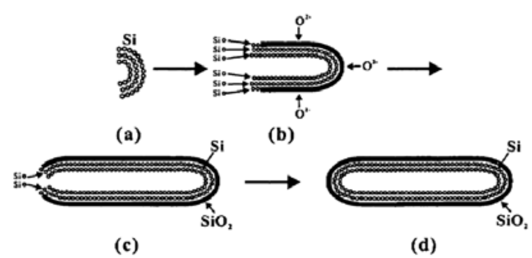
(57) **ABSTRACT**

The present invention relates to a method for preparing self-assembled silicon nanotubes (SiNTs) by a hydrothermal method. A method for preparing self-assembled SiNTs comprises forming a mixture of silicon oxide and water in a sealed container, wherein the mixture has a silicon oxide to water ratio of no more than 10% by weight. The mixture is maintained at a constant temperature and a constant pressure, and the mixture is stirred for a period of time. Self-assembled SiNTs may be formed with an average inner diameter of less than 5 nm and an average outer diameter of around 15 nm. The present invention completely utilizes non-toxic raw materials, and the materials and process do not pollute the environment, so the method satisfies the development trends of the modern industry.

9 Claims, 3 Drawing Sheets



Detailed

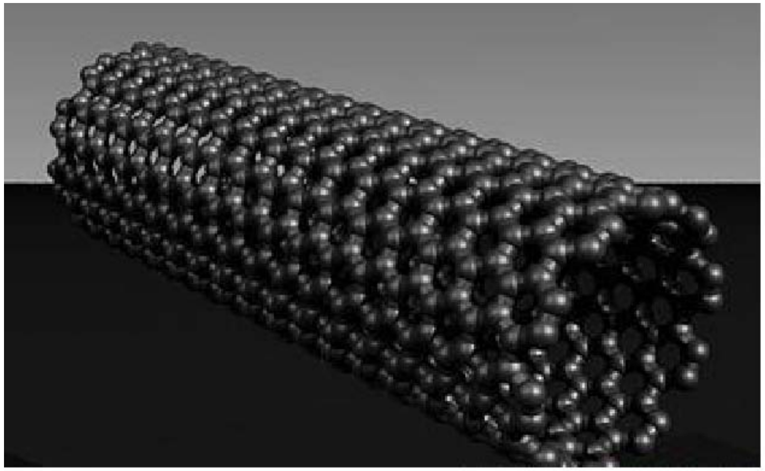
Before using synchrotron lithography

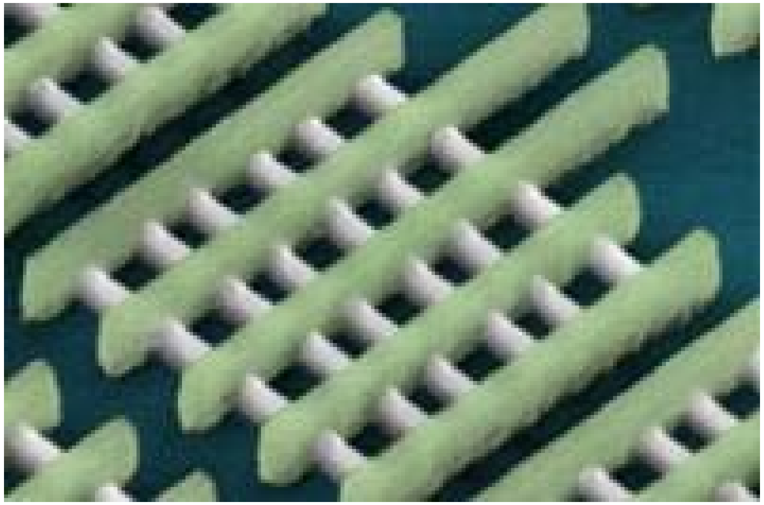
Synchrotron Radiation Center

University of Wisconsin-Madison

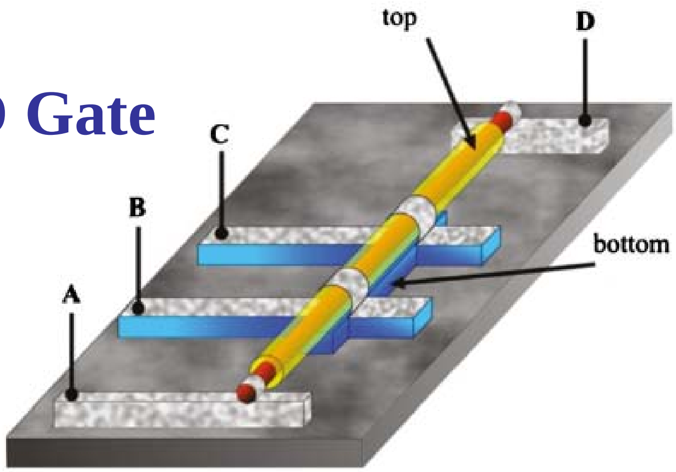
After using synchrotron lithography

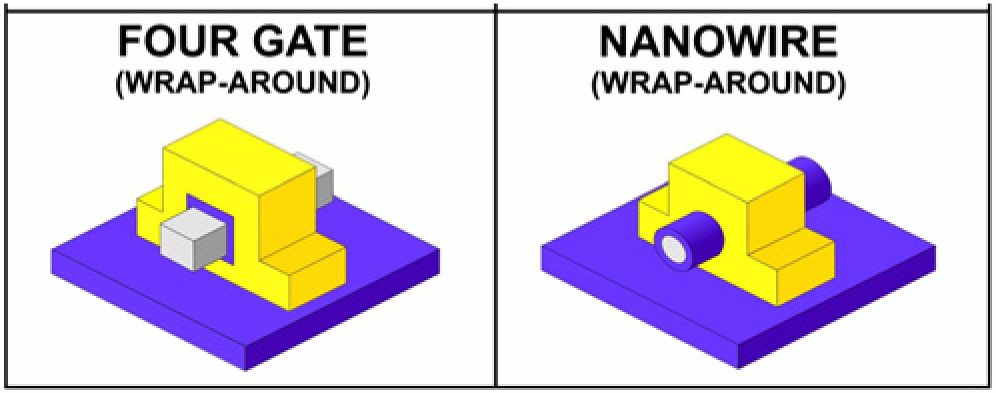
Thermal CVD for synthesis and doping





Dreamed Around-D Gate





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Conclusions

1. This proposal suggests using SR lithography to draw silicon nanotube pattern and then form the base of Around-D Gate of computer chip.
2. Near all common and important nanomaterials, such as C_{60}, carbon nanotube, porous silicon, silicon nanowire, have published in Nature/Science.
3. But there is no silicon nanotube yet.
4. This proposal can make a beautiful pattern to show silicon nanotube has important applications. And also show we can go a little head of Intel Corp.

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